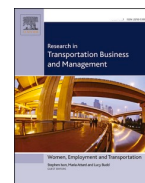




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The equity-environmental sustainability nexus in autonomous vehicles: An integrated framework for policy and planning[☆]

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ABSTRACT

Autonomous Vehicles (AVs) present a dual challenge: their promise of advancing environmental sustainability through efficiency gains is counterbalanced by significant social equity concerns, such as job displacement and unequal access for vulnerable populations. Despite these opportunities and risks, there remains a critical gap in integrated frameworks that optimize both social equity and environmental sustainability in AV deployment. To address this, we conducted a comprehensive literature review, identifying key voids in AV policy, community impacts, behavioral acceptance, and integrated assessment methodologies. Building on these findings, we developed a novel conceptual framework for equitable and sustainable AV deployment. This framework emphasizes the dynamic interplay between AV technology, policy interventions, and their social and environmental outcomes. It is grounded in proactive planning, meaningful community engagement, adaptive governance, and interdisciplinary collaboration, and is designed to be adaptable across diverse deployment contexts. Results provide both theoretical insights and practical recommendations, including an application pathway for iterative policy adaptation. These contributions offer a roadmap for policymakers and planners to guide AV deployment towards mutually reinforcing equitable and sustainable outcomes, ensuring AVs contribute positively to a future that is both green and fair.

1. Introduction

Autonomous Vehicles (AVs) represent a transformative frontier in transportation, promising to revolutionize urban mobility, logistics, and personal travel. Proponents often highlight their potential to enhance environmental sustainability through optimized traffic flow, reduced congestion, increased vehicle utilization, and the accelerated adoption of electric powertrains (Emory, Douma, & Cao, 2022). These advancements are envisioned to lead to significant reductions in greenhouse gas emissions; energy consumption; and urban land dedicated to parking (Kopelias, Demiridi, Vogiatzis, Skabardonis, & Zafiropoulou, 2020). However, alongside these promising environmental benefits; the widespread deployment of AVs introduces complex challenges; particularly concerning social equity. Concerns have emerged regarding potential job displacement for professional drivers; unequal access to AV services; and the exacerbation of existing transportation disparities for vulnerable populations (Fatima, Hsiu Lee, & Dannenberg, 2024). This article posits that the relationship between the environmental sustainability and

social equity impacts of AVs is not merely coincidental but deeply intertwined; forming a critical equity-environmental sustainability nexus. A failure to address this nexus comprehensively risks undermining the very goals of sustainable development. For instance; while AVs could theoretically reduce emissions; if their deployment leads to increased Vehicle Miles Traveled (VMT) due to cheaper; more convenient personal travel; or if they disproportionately benefit affluent communities while burdening marginalized ones with increased traffic or reduced public transit options; the net environmental and social outcomes could be negative (Freemark, Stacy, Fiol, Morales-Burnett, & Weng, 2022).

The overarching research gap addressed in this article is the pervasive lack of integrated policy, planning, and assessment frameworks that simultaneously consider and optimize for both social equity and environmental sustainability in the context of AV deployment. Existing research and policy often treat these two critical dimensions in isolation, leading to fragmented understanding and potentially conflicting outcomes. This siloed approach prevents the development of holistic

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strategies necessary for a truly sustainable and just autonomous mobility future (Bahman, Abahussain, Khan, Mahmood, & Shaker, 2025).

While AVs represent a transformative frontier in transportation, their deployment is shaped by a complex ecosystem that extends beyond equity and environmental sustainability. Key factors including governance arrangements, safety assurance, technology readiness, market structures, and public acceptance, play critical roles in determining AV outcomes across diverse operational and governance dimensions. This article intentionally narrows its focus to the equity–environmental sustainability nexus, acknowledging that other dimensions are essential but outside the scope of the present analysis. The rationale for this boundary is to provide theoretical clarity and depth, while recognizing the broader tensions and uncertainties that characterize AV development and deployment. A balanced overview of these ecosystem challenges is provided below, before the discussion narrows to the integrated equity–environment sustainability lens.

This paper focuses specifically on autonomous passenger transport in urban contexts. This focus reflects both the dominant emphasis of the existing literature and the concentration of current policy experimentation and pilot deployments in cities. While autonomous systems are also emerging in freight and logistics, these domains raise distinct equity and environmental sustainability challenges that fall outside the scope of the present analysis. Narrowing the scope in this way allows for greater conceptual clarity and analytical depth in examining the equity–environment sustainability nexus.

This article aims to address this gap by:

1. Critically reviewing the existing literature on the environmental and social equity impacts of AVs, highlighting the disconnects and specific research voids.
2. Proposing a novel conceptual framework that integrates equity and sustainability considerations into AV policy and planning.
3. Providing concrete policy and planning implications for stakeholders to guide AV deployment towards mutually reinforcing equitable and sustainable outcomes.

By exploring this nexus, this article seeks to make a significant contribution to literature, offering both theoretical insights and practical recommendations for navigating the complex interplay between technological advancement, environmental stewardship, and social justice in the future of transportation.

For clarity and consistency, the following definitions are adopted: *Social equity* refers to the fair distribution of benefits and burdens across all societal groups, ensuring equal access to opportunities and resources regardless of socio-economic status. *Autonomous Vehicles (AVs)* refers to all forms of self-driving vehicles and associated services, including shared and on-demand AVs operating within urban and regional contexts. Throughout the paper, *sustainability* is understood as the simultaneous pursuit of environmental, social, and economic objectives, aligning with the widely recognized three-pillar framework.

The remainder of this paper is organized as follows: [Section 2](#) reviews the relevant literature and identifies key research gaps. [Section 3](#) introduces the proposed integrated framework for equitable and sustainable AV deployment, detailing its core components and policy levers. [Section 4](#) discusses the implications of the framework, including practical pathways for implementation and mechanisms for addressing trade-offs. Finally, [Section 5](#) presents the conclusions, highlighting contributions, limitations, and directions for future research.

2. Literature review

Sustainability is commonly conceptualized as encompassing three interrelated pillars: environmental, social, and economic (Elkington, 1997). These pillars collectively define the scope of sustainable development, with each dimension contributing distinct yet interconnected objectives: environmental protection, social equity, and economic

viability. In the context of AV deployment, these elements often interact in complex ways, shaping both policy priorities and practical outcomes. While this paper focuses primarily on environmental sustainability and social equity, we briefly acknowledge the economic dimension to provide a holistic perspective. Recognizing the economic implications, such as cost efficiency, affordability, and market viability, is essential for understanding the broader impacts of AVs, even though these aspects are not the central focus of our analysis. By foregrounding environmental and social considerations, this article aims to address the most pressing gaps in current research, while remaining mindful of the economic context that underpins sustainable mobility systems (Chandratreya, 2025).

2.1. Autonomous vehicles and environmental sustainability

The environmental implications of AVs are a subject of extensive debate, with projections ranging from significant environmental benefits to potential drawbacks. On the positive side, AVs are anticipated to enhance traffic efficiency through optimized routing, platooning, and reduced human error, leading to smoother traffic flow and decreased fuel consumption and emissions (de Leo & Miragliotta, 2025). The potential for shared autonomous vehicles (SAVs) to reduce private car ownership and increase vehicle utilization could further decrease the total number of vehicles on the road; thereby lowering manufacturing-related emissions and parking infrastructure needs (Dill & McNeil, 2021). Furthermore, the integration of AV technology with electric powertrains is expected to accelerate the decarbonization of the transportation sector; offering substantial reductions in local air pollution and greenhouse gas emissions (Arpaci, Al-Sharafi, & Mahmoud, 2024; Zhong et al., 2023).

However, these optimistic projections are tempered by several potential negative environmental impacts. A primary concern is the possibility of increased VMT. If AV travel becomes significantly cheaper, more convenient, and productive (e.g., allowing occupants to work or relax), it could induce demand for travel, leading to more frequent and longer trips (Duarte & Ratti, 2018). This induced demand could offset or even outweigh the efficiency gains; resulting in higher overall energy consumption and emissions. Additionally, the energy demands of operating AVs; particularly for their sophisticated sensor arrays and computing systems; could be substantial; especially if the energy source is not entirely renewable (Hua, Fan, Hua, & Zhang, 2026; Williams, Das, & Fisher, 2020). Continuous data processing, real-time navigation, and connectivity requirements further increase operational energy consumption, potentially offsetting some environmental benefits if not managed with clean energy sources. The lifecycle environmental impacts, including the manufacturing and disposal of advanced AV components, also warrant careful consideration to ensure a net positive environmental outcome (Martin, 2019). This includes resource extraction for batteries and electronics; emissions from production processes; and challenges related to recycling or safe disposal of complex materials at end-of-life. Comprehensive assessment of these factors is essential to accurately evaluate the sustainability of AV systems (Burchart & Przytuła, 2025).

2.2. Autonomous vehicles and social equity

The social equity implications of AVs are equally complex and present a critical area of concern. AVs hold the promise of enhancing mobility for traditionally underserved populations, such as the elderly, individuals with disabilities, and those unable to drive due to age or economic constraints (Dianin, Ravazzoli, & Hauger, 2021). By providing on-demand; accessible transportation; AVs could significantly improve access to employment; healthcare; education; and social activities; thereby reducing social isolation and enhancing overall quality of life (Dicianno et al., 2021). Conversely, the deployment of AVs also carries substantial risks of exacerbating existing social inequities. Empirical

case-based research from urban contexts further illustrates how autonomous vehicle initiatives can both mitigate and reproduce transportation inequities; depending on governance arrangements; service design; and community engagement practices (Robinson-Tay, 2024).

Zhang et al. (2025) provide insights into jaywalkers' interactions with AVs; highlighting behavioral dynamics that influence safety and operational design. While not focused on equity impacts; these findings underscore the importance of considering pedestrian behavior in AV planning frameworks. This emphasis is echoed in several recent studies; which demonstrate that modeling and understanding pedestrian behavior; including unpredictable actions like jaywalking; that is essential for AV safety and operational planning. For example; Khuzam, Lanzaro, and Sayed (2025) and Muktadir, Jawad, Paranjape, Whitehead, and Shepelev (2022) show that realistic simulation of pedestrian dynamics is critical for assessing AV crash risk and system robustness. Similarly; Rezwana and Lownes (2024) and Oikonomou, Ziakopoulos, and Yannis (2025) highlight that AV planning frameworks must integrate behavioral variability to ensure safe and equitable deployment in real-world urban environments. These findings collectively reinforce the need to incorporate pedestrian behavior into AV design and policy frameworks. Integrating these behavioral insights into AV design and policy frameworks is therefore essential for developing solutions that are truly responsive to the complex social dynamics of urban mobility and contribute to equitable outcomes.

The cost of AV services, especially in early deployment phases, could render them inaccessible to low-income individuals, creating a new form of transportation exclusion. Furthermore, the digital divide poses a barrier, as access to and proficiency with smartphone applications and digital platforms may be necessary to utilize AV services, potentially marginalizing those without such access. The spatial distribution of AV services could also be inequitable, with services initially concentrated in affluent urban areas, leaving underserved communities with limited or no access to these advanced mobility options (Etmiani-Ghasrodashti et al., 2021).

2.3. Disconnects and siloed approaches

A critical observation from the existing literature is the tendency to analyze the environmental sustainability and social equity impacts of AVs as largely separate domains. Research often focuses on optimizing AV technology for environmental performance (e.g., energy efficiency, emissions reduction) without adequately considering the distributional justice of these benefits or the potential for unintended social consequences. Similarly, studies on AVs and equity frequently examine access, affordability, or job impacts without fully integrating the environmental footprint of such deployments (Gao et al., 2025). This siloed approach leads to a fragmented understanding of the overall societal impact of AVs and can result in policies that inadvertently create trade-offs rather than synergies between equity and sustainability goals. For example; a policy promoting widespread private AV ownership for individual convenience might improve personal mobility (an equity aspect) but could drastically increase VMT and energy consumption; thereby harming environmental sustainability (Kovačić, Mutavdžija, & Buntak, 2022). The absence of integrated frameworks means that decision-makers often lack the tools to evaluate the holistic impacts of AV deployment strategies, making it challenging to steer towards outcomes that are simultaneously green and fair.

2.4. Research gaps

Building upon the review of existing literature, several distinct research gaps emerge at the intersection of AVs, environmental sustainability, and social equity. These gaps highlight the need for a more integrated and interdisciplinary approach to understanding and managing the transformative potential of AVs.

There is a significant void in comprehensive policy frameworks that

explicitly integrate social equity and environmental sustainability considerations into AV deployment planning. Existing policies often focus on singular aspects, such as safety or technological regulation, without adequately addressing the complex interplay between these dimensions (Emory et al., 2022; Fatima et al., 2024; Khan et al., 2024). This leads to a reactive rather than proactive approach to managing the societal impacts of AVs. Research is critically needed to develop proactive policy interventions that mandate equitable access to AV services, prevent displacement of vulnerable populations due to AV-driven urban changes, and ensure environmental benefits are distributed fairly across all communities. This includes exploring regulatory mechanisms, incentive structures, and urban planning strategies that foster both sustainable urban forms and equitable access to resources and opportunities (Eppenger & Richter, 2021). The development of ethical frameworks for AV decision-making is also crucial; extending beyond operational safety to encompass broader societal values of sustainability and equity; thereby preventing outcomes that worsen congestion; sprawl; or emissions (Bahman & Naser, 2025; Lopez-Aparicio et al., 2025).

The potential for AVs to either widen or narrow existing transportation disparities for marginalized groups (e.g., low-income individuals, elderly, disabled, rural residents) remains a critical, yet under-researched area. While some studies acknowledge the risk of exacerbating inequities, empirical evidence and prescriptive solutions are often lacking (Tabascio, Tiznado-Aitken, Higgins, & Farber, 2024). Research is needed to investigate how AVs might affect job displacement in the transportation sector, accessibility for non-drivers, the digital divide in accessing AV services, and the potential for increased VMT if AVs lead to cheaper, more convenient personal travel, thereby undermining environmental goals. Furthermore, the environmental implications are profound, as AV deployment patterns could concentrate environmental benefits in affluent areas while continuing to expose vulnerable communities to pollution from legacy transportation systems or increased traffic. Studies focusing on case studies, community engagement models, and impact assessments are essential to understand and mitigate these risks.

Public perception and acceptance are paramount for the successful integration of AVs. However, research often overlooks how perceptions of equity (or inequity) in AV deployment influence public acceptance, especially among diverse socio-economic groups, and how this, in turn, affects the realization of environmental benefits (Hao & Wang, 2023). There is a lack of nuanced understanding regarding how perceptions vary across different demographic and socio-economic groups; with factors like age; income; and prior technological experience significantly influencing acceptance. Research should explore community-specific concerns and needs regarding AVs; the role of trust and transparency in AV deployment; and how to design AV services that are perceived as fair; accessible; and environmentally responsible by all segments of society. The application of behavioral economics to encourage equitable and sustainable AV usage patterns; such as incentivizing shared AV services and discouraging unnecessary private AV ownership; also represents a significant research gap (Shen, Wang, & Gifford, 2021).

A recurring challenge in sustainable transport research is the lack of comprehensive data and integrated methodologies, particularly when linking environmental outcomes with social equity metrics in the context of emerging technologies like AVs. Parekh et al. (2022) conducted an extensive synthesis of AV technologies, systematically analyzing 74 studies to map the current landscape of evaluation methodologies. Their review reveals that most existing approaches are fragmented, with technical assessments often isolated from broader social and environmental considerations. The authors highlight that the lack of standardized metrics and interoperable data systems impedes meaningful cross-study comparisons and limits the ability to evaluate AV impacts in a comprehensive manner. They further emphasize that current frameworks rarely integrate equity and sustainability dimensions, resulting in assessments that overlook critical factors such as

accessibility, distributional justice, and long-term ecological outcomes. By advocating for holistic, interoperable evaluation models, the authors underscore the necessity of developing integrated metrics that can capture the multifaceted effects of AV deployment, thereby supporting the case for frameworks that address both equity and environmental sustainability in tandem.

There is a critical shortage of comprehensive datasets that simultaneously capture environmental metrics (e.g., emissions, energy consumption, land use changes) and social equity indicators (e.g., accessibility, affordability, health outcomes, job impacts) related to AV deployment (Wu, Cao, & Douma, 2021). Novel data collection methods and integrated assessment frameworks are needed to evaluate the environmental footprint; and social equity impacts of AV systems concurrently. Methodological challenges include developing appropriate metrics for accessibility; affordability; and distributional justice; and integrating these with environmental impact models. Furthermore; there is a need for more sophisticated dynamic modeling and scenario analysis tools that can project the long-term; interconnected impacts of different AV deployment strategies on both equity and sustainability (Tabascio et al., 2024).

3. Towards an integrated framework for equitable and sustainable AV deployment

This paper adopts a conceptual and interpretive approach rather than a systematic or integrative review. Articles were selected to reflect influential policy debates, empirical insights, and conceptual contributions relating to autonomous passenger transport, equity, and environmental sustainability. The analysis involved thematic synthesis across these strands to identify recurring gaps, tensions, and integration challenges that informed the development of the proposed framework.

3.1. Rationale for integration

The preceding literature review unequivocally demonstrates that a fragmented approach to AV deployment, which addresses environmental sustainability and social equity in isolation, is insufficient and potentially counterproductive. The complex interplay between technological advancements, urban development, and societal well-being necessitates a holistic perspective. The rationale for an integrated framework is rooted in the understanding that environmental degradation disproportionately affects marginalized communities, and conversely, social inequities can hinder the adoption and effectiveness of sustainable solutions (Gavanas, Anastasiadou, Nathanail, & Basbas, 2024; Othman, 2022). Therefore, achieving truly sustainable autonomous mobility requires a framework that recognizes and actively manages the interdependencies and potential trade-offs between equity and sustainability goals, striving for synergistic outcomes where possible. Such an integrated approach moves beyond simply mitigating negative impacts to proactively designing AV systems that contribute positively to both ecological health and social justice (Khan et al., 2024; Naser, Bahman, & Shaker, 2025).

3.2. Proposed conceptual framework

To address the identified research gaps and foster a more holistic understanding, we propose a conceptual framework for equitable and environmentally sustainable AV deployment. This framework (Fig. 1) illustrates the dynamic relationships between AV technology, policy and planning interventions, and their resultant social and environmental outcomes, emphasizing feedback loops that inform continuous adaptation and improvement. Although the framework emphasizes environmental sustainability and equity, it recognizes that economic considerations, such as cost efficiency and affordability, are integral to comprehensive AV planning.

The framework is built upon three core components:

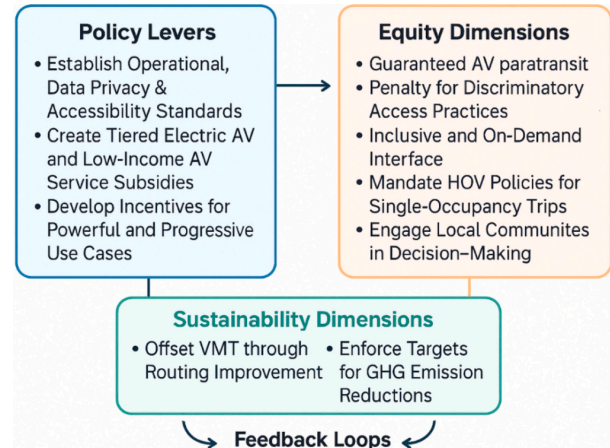


Fig. 1. Conceptual Framework for Equitable and Sustainable AV Deployment.

- 1. Policy Levers:** These represent deliberate actions by governments and planners to shape AV deployment. Key levers include regulation, incentives, infrastructure investment, and urban planning strategies.
- 2. Equity Dimensions:** These capture social objectives such as guaranteed accessibility, affordability, inclusive interfaces, and community participation in decision-making.
- 3. Sustainability Dimensions:** These focus on environmental goals, including reducing VMT, enforcing greenhouse gas emission targets, and promoting resource efficiency.

Feedback loops connect these elements, ensuring continuous adaptation and improvement based on observed outcomes.

While Fig. 1 visually emphasizes the policy levers, equity dimensions, and sustainability dimensions, the framework recognizes that AV system design and technology choices are foundational and interact with these components. Technological characteristics, such as automation level, propulsion system, and service model, both shape and are shaped by policy interventions and are reflected in equity and sustainability outcomes.

Fig. 1 presents a conceptual framework for integrating equity and environmental sustainability considerations in urban autonomous passenger transport systems. The framework is grounded in policy and planning dynamics typically observed in cities, where passenger mobility, land use, and public transport integration are central concerns. While the structure of the framework is conceptually transferable, its specific equity dimensions, sustainability objectives, and policy levers are intentionally framed around urban passenger applications. This means that other applications, such as freight and logistics, first/last-mile services, remote-area operations, and controlled environments, each involve distinct risks, infrastructure needs, and business constraints. By making these assumptions explicit, we aim to strengthen the conceptual foundations underpinning the framework, while acknowledging that future research may adapt its structure to other AV contexts.

The framework is designed to be adaptable: the four policy levers can be generalized and tailored to address the specific requirements and challenges of these diverse settings. For example, infrastructure investment may prioritize connectivity and reliability in freight corridors, while urban planning may focus on integration with existing logistics hubs or remote-area accessibility. By clarifying these assumptions, we aim to clarify the conceptual assumptions underpinning the framework, while acknowledging that future research may adapt its structure to other AV contexts.

The four policy levers provide the foundation for shaping equitable and sustainable outcomes within the urban autonomous passenger transport contexts examined in this study. *Regulation* establishes clear

operational, privacy, and accessibility standards to ensure safe and fair integration of AVs into existing mobility systems. *Incentives*, both financial and non-financial, are essential to promote sustainable AV choices, such as shared and electric vehicles, while also supporting vulnerable and underserved populations through targeted subsidies or service provisions. *Infrastructure* investment plays a critical role in enabling multimodal integration, supporting efficient AV operation, and minimizing environmental impacts. Complementing this, *urban planning* must proactively guide land-use patterns to prevent AV-induced sprawl, foster compact development, and strengthen connections with public transport systems. The adaptation of High-Occupancy Vehicle (HOV) lanes to prioritize shared autonomous vehicles and the integration of AV paratransit services (on-demand, shared AVs) are crucial. These initiatives optimize urban mobility, reduce congestion, and enhance equitable access to transportation, thereby directly supporting sustainable and inclusive urban development.

The equity dimension of the framework focuses on ensuring fair access, affordability, participation, and the equitable distribution of benefits and burdens. This entails guaranteeing both physical and digital access to AV services for all groups, including individuals with disabilities, the elderly, and low-income communities. Economic accessibility is addressed through mechanisms such as tiered pricing structures, public-private partnerships, and targeted subsidies. Meaningful participation of marginalized communities in AV planning and governance is essential to ensure that deployment strategies reflect diverse needs and social contexts. Moreover, the framework emphasizes the fair distribution of AV-related advantages, such as reduced emissions and enhanced mobility, while mitigating potential disadvantages like job displacement or localized traffic increases.

From a sustainability perspective, the framework integrates goals related to emissions reduction, energy efficiency, land use, and resource circularity. Priorities include minimizing greenhouse gas emissions through electric and efficiently routed AVs, improving the energy performance of vehicles and infrastructure through renewable energy integration, and promoting compact, efficient land use by reducing parking demand and reclaiming urban space for community and ecological purposes. Furthermore, resource efficiency is achieved by embedding circular economy principles throughout the AV life cycle, from manufacturing and operation to maintenance and end-of-life management.

Finally, the framework highlights feedback loops between assessment outcomes and decision-making processes. Insights derived from social and environmental evaluations are intended to continuously inform AV system design, technology development, and policy refinement. This adaptive governance approach fosters learning and iterative improvement, ensuring that AV deployment evolves in alignment with societal values, equity goals, and long-term sustainability objectives.

3.3. Key principles for integrated planning

Effective implementation of the proposed framework necessitates adherence to several key principles for integrated planning. Effective management of the equity-environment sustainability nexus in AVs necessitates adherence to several key principles. Firstly, proactive and anticipatory planning is crucial, moving beyond reactive measures to integrate potential equity and sustainability challenges and opportunities from the earliest stages of policy and infrastructure development (Ma, Zhao, Wang, Li, & Zhou, 2024). Secondly, meaningful community engagement is paramount; ensuring active and inclusive participation from diverse, especially historically marginalized, communities. This approach fosters co-creation of AV solutions that meet local needs and reflect community values; thereby building trust and acceptance (Bradley & Mahmoud, 2024). Thirdly, adaptive governance is essential; given the rapid technological advancements and inherent uncertainties of AV deployment. Policies must be flexible; incorporating mechanisms for regular review; evaluation; and adjustment based on real-world

outcomes and emerging data (Alghodhaifi & Lakshmanan, 2021). Fourthly, interdisciplinary collaboration is indispensable, requiring experts from transportation engineering, urban planning, environmental science, sociology, economics, and public policy to work together across traditional disciplinary boundaries. Finally, data-driven decision-making with integrated metrics is vital, necessitating robust data collection and the development of metrics that capture both social equity and environmental sustainability outcomes. This enables evidence-based policy formulation and transparent evaluation of AV projects. It is essential that policy ambition is grounded in the practical realities of technology and infrastructure readiness, so that equity and sustainability objectives remain actionable as AV systems and supporting environments mature.

4. Discussion and implications

The proposed integrated framework for equitable and sustainable AV deployment directly addresses the research gaps identified in Section 2.4 by providing a structured approach to bridge the current disconnects between equity and sustainability considerations. By emphasizing the interdependencies between AV technology, policy interventions, and their social and environmental outcomes, the framework encourages a holistic perspective that moves beyond siloed analyses. Specifically, the framework advocates for proactive, integrated policy interventions that explicitly consider both equity and sustainability from the outset. This contrasts with current reactive and fragmented approaches, promoting the development of regulatory mechanisms, incentive structures, and urban planning strategies that are designed to achieve dual objectives. The inclusion of ethical frameworks within policy levers ensures that AV deployment aligns with broader societal values, preventing negative externalities such as increased congestion or sprawl (Borenstein, Bucher, & Herkert, 2022; Schepis, Purchase, Oлару, Smith, & Ellis, 2023). By explicitly incorporating equity dimensions such as access, affordability, participation, and the fair distribution of benefits and burdens, the framework compels policymakers and planners to consider the specific needs and potential vulnerabilities of marginalized groups. This proactive consideration aims to mitigate risks like job displacement, address the digital divide, and ensure that environmental benefits are shared equitably, thereby advancing environmental justice (Ulibarri, Figueroa, & Grant, 2022).

Also, the framework recognizes that public acceptance is not monolithic but varies across socio-economic groups. By emphasizing meaningful community engagement and understanding heterogeneous perceptions, it guides the design of AV services that are perceived as fair, accessible, and environmentally responsible. This approach helps to build trust and foster the behavioral changes necessary for sustainable AV adoption, such as increased use of shared mobility options (Garus, Mourtzouchou, Suarez, Fontaras, & Ciuffo, 2024). In addition, the framework underscores the critical need for novel data collection methods and integrated assessment tools that can simultaneously evaluate both environmental and social equity impacts. By advocating for interdisciplinary collaboration and dynamic modeling, it pushes for the development of comprehensive metrics and analytical approaches that can capture the complex interactions and trade-offs, moving beyond descriptive analyses to inform predictive and prescriptive policy.

The integrated framework offers several concrete policy implications for various levels of governance. First, governments should develop overarching policy mandates that require integrated equity and sustainability assessments for all major urban passenger AV deployment initiatives. This includes setting clear targets for emissions reductions and equitable access and establishing funding mechanisms that prioritize projects demonstrating dual benefits. Policies should also address potential job displacement through workforce retraining programs and social safety nets. Second, city planners must integrate AV planning into broader urban development strategies, using land-use policies to prevent sprawl and promote compact, mixed-use developments that

support multimodal transportation. This involves zoning regulations that encourage shared AV services over private ownership and repurpose parking spaces for public amenities or green infrastructure. Community engagement should be a cornerstone of local AV planning, ensuring that solutions are tailored to local needs and priorities. In addition, regulators need to establish clear standards for AV accessibility, data privacy, and ethical decision-making. This includes developing mechanisms to monitor and enforce equitable service provision and environmental performance, potentially through performance-based regulations that incentivize positive social and environmental outcomes. This is crucial as policymakers should design incentives that encourage the adoption of electric, shared, and accessible AVs. This could include tax credits for shared AV fleet operators, subsidies for low-income users of AV services, and preferential treatment for AVs that demonstrate superior environmental and equity performance.

For urban planners and developers, the integrated framework suggests a shift towards more holistic and anticipatory planning practices. Future transportation infrastructure should be designed to accommodate shared and passenger-oriented AV services while prioritizing pedestrians, cyclists, and public transit. This includes dedicated AV lanes for shared services, smart infrastructure that communicates with AVs to optimize traffic flow and charging infrastructure for electric AVs that is equitably distributed across urban areas. Planners should leverage AVs to reduce the demand for parking, freeing up valuable urban land for housing, green spaces, or other community assets. This requires rethinking traditional zoning and development regulations to encourage higher density, mixed-use developments that are well-served by shared AVs and public transit. Planners need to invest in advanced data collection and modeling capabilities to understand the complex interactions between AVs, urban form, and social and environmental outcomes. This includes developing simulation models that can test different AV deployment scenarios and their impact on VMT, emissions, accessibility, and equity indicators. Planning processes must therefore be community-centric, involving residents in the design and implementation of AV services. This ensures that AV solutions are responsive to local needs, address specific mobility challenges, and contribute to the overall well-being of diverse neighborhoods (Moharam & Mukherjee, 2024).

The ethical imperative of prioritizing equity and sustainability in AV development and deployment cannot be overstated. The framework reinforces that technological advancement, while powerful, is not inherently benevolent. Without explicit ethical guidelines and robust governance, AVs could exacerbate existing societal problems. Ethical considerations must move beyond accident liability to encompass the broader societal impacts, including the fair distribution of benefits and burdens, the protection of vulnerable populations, and the stewardship of the environment. This requires ongoing public discourse, transparent decision-making, and accountability mechanisms to ensure that AVs serve the public good (Mukherjee, Sangal, Sarkar, Alkadash, & Almamari, 2023; Soh & Martens, 2024).

While the integrated framework aims to foster synergies between equity and sustainability, it is crucial to acknowledge that potential trade-offs may exist. For example, optimizing AV routes for maximum environmental efficiency might inadvertently bypass certain low-demand areas, impacting equitable access. Conversely, mandating universal AV service in all areas, regardless of demand, could increase operational costs and potentially lead to higher energy consumption. The framework encourages a nuanced approach to identify and manage these trade-offs through:

- **Multi-objective Optimization:** Developing decision-making tools that can optimize for multiple objectives (e.g., minimizing emissions while maximizing accessibility) and explore the Pareto frontier of possible solutions.
- **Context-Specific Solutions:** Recognizing that the optimal balance between equity and sustainability may vary depending on local contexts, urban forms, and socio-economic conditions.
- **Adaptive Management:** Implementing policies with built-in flexibility to adjust strategies as real-world data on trade-offs and synergies emerge.

Ultimately, the goal is to identify and maximize synergies, such as shared electric AVs that reduce both emissions and transportation costs for low-income users, while proactively mitigating unavoidable trade-offs through compensatory measures or alternative solutions. To operationalize these principles and provide practical guidance, Fig. 2 outlines an application pathway that integrates stakeholder engagement, policy adaptation, and iterative learning mechanisms. Policymakers should follow an iterative process that combines technical readiness with community engagement. After defining equity and sustainability goals, stakeholder involvement and trust-building ensure inclusivity and legitimacy. Policy levers are adapted to local contexts where pilot programs and impact assessments provide real-world evidence, enabling refinement and scaling. This approach embeds experiential learning and adaptive management, ensuring AV deployment aligns with societal values and sustainability objectives.

The application pathway illustrated below demonstrates how the framework may be operationalized in urban passenger transport contexts, where policy levers, stakeholder engagement, and sustainability objectives are most clearly defined.

Application Pathway for the Integrated Equity-Sustainability in AV Deployment

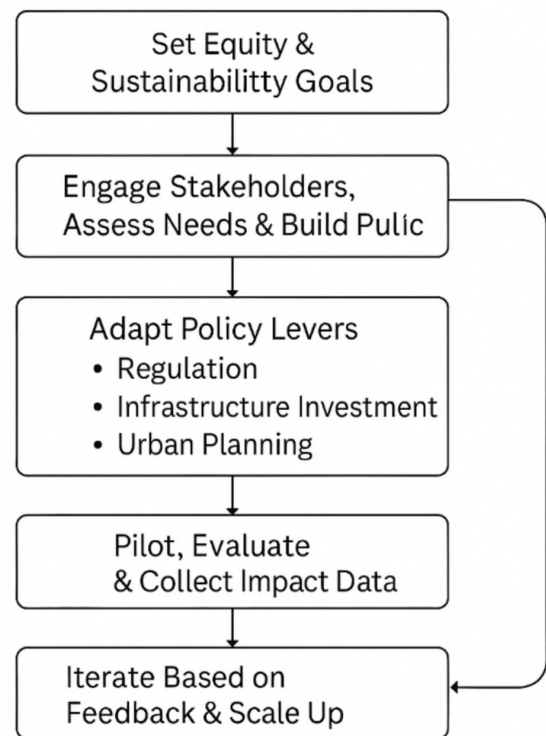


Fig. 2. Application Pathway for Operationalizing the Integrated Equity-Sustainability Framework in AV Deployment.

5. Conclusion

This article has explored the critical equity-environmental sustainability nexus in urban autonomous passenger transport, arguing that an integrated policy and planning approach is essential for realizing the potential benefits of AVs while safeguarding environmental outcomes and social equity. We have demonstrated that current research and policy often treat equity and sustainability as separate concerns, leading to fragmented understanding and the risk of unintended negative consequences. The comprehensive literature review highlighted significant research gaps across policy and planning, impacts on vulnerable communities, behavioral acceptance, and integrated assessment methodologies.

To address these gaps, we proposed a conceptual framework for equitable and sustainable AV deployment in urban passenger transport contexts, emphasizing the dynamic interplay between AV technology, policy interventions, and social and environmental outcomes. Grounded in proactive planning, meaningful community engagement, adaptive governance, and interdisciplinary collaboration, the framework provides structured guidance for policy and planning in cities where passenger mobility, land-use pressures, and equity concerns intersect. Distinctively, the framework incorporates an application pathway that enables iterative policy adaptation, stakeholder engagement, and continuous learning, ensuring that AV deployment remains responsive to evolving societal needs and technological advancements. While the conceptual logic of the framework may inform broader AV debates, the analysis, examples, and policy implications presented here are intentionally confined to urban autonomous passenger transport and should not be assumed to directly translate to freight or logistics applications without further empirical validation.

This article contributes significantly to the existing literature by providing a much-needed integrated perspective on the equity-sustainability nexus in autonomous mobility. By synthesizing disparate research streams and identifying critical voids, it lays the groundwork for a more coherent and responsible approach to AV development and deployment. The proposed framework and principles offer a theoretical foundation and practical guidance for policymakers, urban planners, and researchers seeking to navigate the complexities of this emerging technological landscape. Importantly, our analysis intentionally focuses on the equity-environmental sustainability nexus within urban passenger autonomous mobility, while recognizing that economic and other dimensions remain important areas for future research.

Future research should focus on empirical validation of the proposed framework, including case studies of AV pilot programs that explicitly integrate equity and sustainability metrics. There is a pressing need for methodological advancements in data collection and integrated assessment tools that can accurately quantify the complex social and environmental impacts of AVs across diverse contexts. Further exploration into the behavioral economics of AV adoption, particularly among marginalized groups, and the development of participatory planning models will also be crucial. Additionally, advancing adaptive management strategies and robust feedback mechanisms will be vital to ensure that AV deployment evolves in alignment with societal values, equity goals, and long-term sustainability objectives. Ultimately, by continuously refining our understanding and actively working towards integrated solutions, we can ensure that AVs contribute to a future that is truly sustainable and equitable for all.

Use of Artificial Intelligence (AI)

This manuscript was prepared by the authors with the assistance of AI-based writing support tools for language and editorial clarity. All analysis, interpretation, and conclusions are original.

CRediT authorship contribution statement

Nahed Bahman: Visualization, Project administration, Investigation, Formal analysis, Data curation, Writing – review & editing, Writing – original draft. **Naser Naser:** Visualization, Conceptualization, Writing – original draft. **Tariq Mahmood:** Supervision, Writing – review & editing, Writing – original draft.

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Declaration of competing interest

The authors declare that there is no conflict of interest relevant to the content of this manuscript.

Data availability

No data was used for the research described in the article.

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